

RENANCE

FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

CHM 101 — Past Questions & Materials

by Renance

Course: CHM 101

Institution: Federal University of Technology, Minna

Papers in this pack: 1

Levels: 100

This study pack was compiled and restructured by Renance for CHM 101 students at Federal University of Technology, Minna. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — CHM101 PQ

Level 100 • First Semester • PastQuestion

According to the kinetic theory, molecules of different gasses at the same temperature always have the same ____

Average density

Volume

Pressure

Average kinetic energy

Doubling the volume of an ideal gas at constant temperature will

Double the pressure

Half the temperature

Half the pressure

Increase the pressure by halve

As solid elements melts, the atom becomes ____ and they have ____ attraction for one another

Closer together, more

More separated, more

Larger greater

More separated, less

The tendency of an electrode to release to or attract electron from the surrounding electrolyte solution at 298K temperature 1atm pressure and 1ml/dm conc. Is known as

3

Electrolytic potential

Electrode potential

Standard electrolytic potential

Standard electrode potential

Consider the reaction $2\text{AgNO}_3 + \text{H}_2\text{S} \rightleftharpoons \text{Ag}_2\text{S} + 2\text{HNO}_3$;

3 2 (g) 2 (s) 3(aq)

Which of the following affects the value of the solubility product?

Increase in temperature

The addition of aqueous sodium sulphate

Addition of aqueous silver nitrate

The presence of excess acid

The molar solubility of lead (II) chloride is 1.5×10^{-2} mol/dm³ . What is

-2 3

the solubility product of the salt?

3.0×10^{-2}

3.0×10^{-4}

1.5×10^{-2}

1.4×10^{-5}

When ΔH is negative, the reaction is said to be

Endothermic

Exothermic

Reversible

Catalytic

Which of the following statement is true of an acid conjugate base?

Neutralized one another

Differ only by proton

Form salt and water only

Carry opposite charges

The pK of the base XOH , YOH and ZOH are 2.70, 8.60 and 4.50

b

respectively. Arrange them in order of increasing strength

XOH , YOH , ZOH

YOH , XOH , ZOH

XOH , ZOH , YOH

YOH , ZOH , XOH

Use the data below to answer question 41 - 43

The following are the solubility product of some sulphides in mol/dm^3

$\text{MnS} = 2.5 \times 10^{-10}$, (ii) $\text{FeS} = 1.0 \times 10^{-24}$ (iii) $\text{ZnS} = 4.5 \times 10^{-24}$ (iv) $\text{PbS} =$

1.3×10^{-19}

0.01 mol/dm^3

Give that $1.3 \times 10^{-3} \text{ mol/dm}^3$ of hydrogen sulphide is bubbled through a

0.01 mol/dm^3

solution containing 0.01 mol/dm^3 each of the respective cations.

3

The order in which the salt will precipitate is

MnS , FeS , ZnS , PbS

FeS , MnS , ZnS , PbS

PbS , ZnS , FeS , MnS

ZnS , FeS , PbS , MnS

The cation with the lowest concentration is

Zn^{2+}

Pb^{2+}

Fe^{2+}

Mn^{2+}

The cation with the highest concentration is

Zn $2+$

Pb $2+$

Fe $2+$

Mn $2+$

The ionic product in a saturated aqueous solution at a given temperature is given as

K_n

K_w

K_{sp}

K_a

When POH is less than 7.0 the solution is said to be

Neutral

Alkaline

Acidic

Saturation

Which of the following solutions is buffer system?

$\text{KNO}_3 / \text{HNO}_2$

$\text{HCOOK} / \text{HCOOH}$

$\text{KH}_2\text{PO}_4 / \text{H}_2\text{PO}_4^-$

$\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$

$\text{KHSO}_4 / \text{H}_2\text{SO}_4$

0.001mol/dm of Ba(OH) has a pH of

3

2

7

11

9

10

Calculate the hydrolysis constant of 0.1m NH₄Cl solution in 1dm of

4

3

water (K_a of NH₄⁺ is 1.8×10^{-5} mol/dm³)

10^{-3}

5×10^{-3}

5.6×10^{-9}

5.6×10^{-10}

5.6×10^{-20}

5.6×10^{-21}

Which of the statements best, describes an anode in an electrochemical cell?

The electrode, at which oxidation occurs

The electrode at which electrons enter the cell

The electrode at which reduction occurs

The electrode at which higher deposition occurs

The pH of 0.01 HCl solutions is?

1.0

2.0

3.0

4.0

The POH of 0.01 NaOH solutions is

10

8

12

4

2

Which of the following can act as bronsted acid?

BF₃

HS

NH₃

CO₂

CO₃²⁻

Lowry Bronsted defines acid as

A proton acceptor

An electron acceptor

A proton donor

An electron donor

Calculate the concentration of OH⁻ ion in a HCl solution whose

H⁺ is 1.3mol/dm³

+ 3

7.7×10^{-17}

7.7×10^{-16}

7.7×10^{-15}

7.7×10^{-14}

Calculate the pH of a solution whose hydrogen ion concentration is

0.76mol/dm³

1.0

2.0

0.12

1.2

When a weak acid react with a strong base in a neutralization reaction, the resultant solution is

Basic

Neutral

Acidic

Normal

The weakest interparticle attraction exist between particles of ____ and the strongest interparticle attraction exist between particles of a

Liquid, gas

Gas, liquid

Solid, gas

Gas, solid

Which of the following statement is not true about the effect of an increase in temperature on the distribution of molecular speeds in a gas?

The most probable speed increase

The fraction of the molecules with the most probable speed increase

The distribution becomes broader

The area under the distribution curve remains the same as under the lower temperature.

A reaction involving two different reactants can never be

Unimolecular reaction

First order reaction

Second order reaction

Enmolecular reaction

Energy conservation is the basis of

Avogadro's hypothesis

Hess's law

Graham law of diffusion

Law of multiple proportion

The entropy of a system will increase in the order

Gas > liquid > solid

Liquid > solid > gas

Solid > liquid > gas

None of the above

The property of substance that relate to its ability to form new substances are labeled

Physical

Chemical

Extensive

Intensive

Catenation

An example of physical change produce by addition of energy is

Lighting of a match

Burning

Melting of ice

Condensation of steam

When a substance change and still retains its original properties, the change is called ____

Physical change

Endothermic change

Exothermic change

Chemical change

Which of the following is true? If two gases under the same conditions have the same number of molecule then they must (i) have the same number of molecule (ii) occupy the same volume (iii) different number of molecules (iv) occupy different volume

I and II

II and III

III and IV

In thermo chemistry, the reference standard state for gases, liquid and solid is said to be at

25 C/1atm

o

0 C/10atm

o

1 C/60atm

o

1273K/10atm

The thermodynamic state function that is a measure of disorderliness is

Lavoisier

Dalton

Proust

Boyle

The salt resulting from the reaction of both strong acid and base is likely to be a/an

Acid salt

Basic salt

Neutral

No reaction

Use this question to answer the following next two questions

If the hydrogen ion of a solution is given as 0.0001 mol/dm^3 what is the

3

pH of the solution?

1

2

3

4

8

From the result obtain above we can conclude that the solution is

Acidic

Basic

Neutral

Natural

The pK of the base XOH , YOH and ZOH are 2.70, 8.60 and 4.50

b

respectively. Arrange them in order of increasing strength

XOH , YOH , ZOH

YOH , XOH , ZOH

XOH , ZOH , YOH

YOH , ZOH , XOH

Lowry Bronsted defines acid as

A proton acceptor

An electron acceptor

A proton donor

An electron donor

Doubling the volume of an ideal gas at constant temperature will

Double the pressure

Half the temperature

Half the pressure

Increase the pressure by halve

When a strong acid react with a weak base in a neutralization reaction, the resultant solution is

Basic

Neutral

Acidic

Normal

When a strong acid react with a strong base in a neutralization reaction, the resultant solution is

Basic

Neutral

Acidic

Normal

At equilibrium, the change in free energy of a reaction is

Positive

Zero

Minimum

Negative

The Free Energy of a Standard Solution at equilibrium is

Positive

Zero

Minimum

Negative

The unit of measurement for the solubility of a substance is_____

mol ⁻¹

mol dm ³

mol dm ⁻³

dm ³ /mol

3

Obtain K_c for the reaction $\text{H}_2 + \text{O}_2 \rightleftharpoons \text{H}_2\text{O}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

$\frac{[\text{H}_2][\text{O}_2]}{[\text{H}_2\text{O}]^2}$

None of the Above

FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

COURSE- CHM 111

BY CLEVER-B TEAM

1. A chemical reaction is said to be at equilibrium at. A: Minimal degree of freedom and maximum entropy B: Minimal degree of freedom and minimal entropy C: Maximum degree of freedom and maximum entropy

2. Predict the spontaneity of a chemical process if $\Delta G^\circ = 141.7$ and $K = 1.4 \times 10^{-25}$ at 298K. A: Spontaneous process B: Non spontaneous C: An equilibrium process

3. Estimate the boiling point of water given $\Delta H^\circ = 44.01 \text{ kJ mol}^{-1}$

and $\Delta S^\circ = 118.3 \text{ J/Kmol}^{-1}$. A: 100°C B:

93.7°C C: 97.3°C

4. Which of the following statements is true for the effect of catalyst on equilibrium. A: affects the rates of the reactions B: alter the equilibrium position C: no effect on rates and equilibrium position

5. Which of the following principles is used in predicting changes in equilibrium concentrations? A: Le Chateliers B: Zeroth C: Boyles

6. The following reaction is at equilibrium. $\text{C}_{12}(\text{g}) + 3\text{F}_2(\text{g}) \rightleftharpoons$

$2\text{CIF}_3(\text{g})$ How will the system respond if the volume is increased at constant temperature? A: The reaction will shift to the right. B: There will be no change to the equilibrium position C: The reaction will shift to the left

7. The following reaction is at equilibrium. $\text{CF}_2\text{Br}_2(\text{g}) \rightleftharpoons \text{CF}_2(\text{g}) + 2\text{Br}(\text{g})$ $\Delta H = 424 \text{ kJ mol}^{-1}$ How will the system respond if the temperature is decreased? A: The reaction will shift to the left. B: The reaction will shift to the right. C: reaction stops

8. Consider this equation: $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}_2(\text{g})$ Suppose the equation is rewritten as $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g})$ with an equilibrium constant K_c' . What is the relationship between K_c and K_c' ? A:

$K_c^\circ = K_c$ B: $K_c^\circ = (K_c)^{1/2}$ C: $K_c^\circ = \frac{1}{2}(K_c)$

9. Identify the INCORRECT statement below regarding chemical equilibrium. A: All chemical reactions are, in principle, reversible. B: Equilibrium is achieved when the concentrations of species become constant. C: Equilibrium is achieved when reactant and product concentrations are equal.

10. In which of the following reactions will the point of equilibrium shift to the left when the pressure on the system is increased? A: $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g})$ B: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ C: $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{MgO}(\text{s})$

11. What happens when a catalyst is added to a system at equilibrium? A: The reaction follows an alternative pathway of lower activation energy. B: The heat of reaction decreases. C: The potential energy of the reactants decreases.

12. The reaction $\text{A} \rightleftharpoons \text{B}$ has an equilibrium constant of $K = 10^{-4}$. Which of the following statements is always correct? A: The reaction will have 50% product B and 50% reactant A at equilibrium. B: The reaction is very favourable and will have mostly product B at equilibrium. C: The reaction is unfavourable and will not have very much product B at equilibrium.

13. Which of the following statements most accurately relates the properties of a liquid at room temperature with its vapour pressure? A: liquid low vapour pressure will probably have a high surface tension and a high boiling point. B: liquid low vapour pressure will probably have a low surface tension and a high boiling point. C: liquid high vapour pressure will probably have a low surface tension and a high boiling point.

14. For a reversible reaction, the equilibrium lies to the. A: Left B: Right C: middle
15. Which of the following is true for the composition of equilibrium mixture. If $\Delta G = 0$ and $K = 1$ the mixture is. A: Mostly products B: Mostly reactants C: Neither reactants nor products are favoured
16. Calculate the value of K at 298K for the following reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ Given $\Delta G = -32.96 \text{ kJ mol}^{-1}$. A: 5.97×10^5 B: 13.3 C: -13.3
17. Of the following reactions, which of the reaction process is only spontaneous at high temperatures? A: $\Delta H +$, $\Delta S -$ B: $\Delta H +$, $\Delta S +$ C: $\Delta H -$, $\Delta S -$
18. The incomplete combustion of carbon is described by the following equation $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$ How does the spontaneity of this process depend upon temperature? A: spontaneous ($\Delta G < 0$) at all temperatures. B: Spontaneous process at high temperature C: Non spontaneous process
19. A condition of a predation process describes an endothermic process with an increase in system entropy, ΔG will be negative if. A: $T\Delta S$ is greater than ΔH B: $T\Delta S$ is less than ΔH C: $T\Delta S = 0$
20. At 25°C, a reaction has a Gibbs free energy change of +45kJ. If the enthalpy change of the reaction is +35kJ, what is the entropy change of the reaction? A: -400JK B: -33.6JK C: 33.6JK
21. Energy can neither be created nor destroyed but can be converted from one form to other is inferred from. A: zeroth law of thermodynamic B: first law of thermodynamics C: second law to thermodynamics
22. From the equation $2NO_2(g) \rightleftharpoons N_2O_4(g)$ When the value of the reaction quotient before any reaction occurs is zero (0) at 25 °C, the concentration changes so that at equilibrium, $[NO_2] = 0.016 \text{ M}$ and $[N_2O_4] = 0.042 \text{ M}$. What is the value of the equilibrium constant for the reaction? A: 1.6×10^2 B: -1.6×10^5 C: 0.016
23. For the reaction, $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ the concentrations at equilibrium are $[SO_2] = 0.90 \text{ M}$, $[O_2] = 0.35 \text{ M}$, and $[SO_3] = 1.1 \text{ M}$. What is the value of the equilibrium constant, K_c ? A: $K_c = 3.4$ B: $K_c = 6.9$ C: $K_c = 4.3$
24. A 1.00-L flask containing 0.0500 mol of $NO(g)$, 0.0155 mol of $Cl_2(g)$, and 0.500 mol of $NOCl$ $2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$ $K_c = 4.6 \times 10^4$ Calculate the reaction quotient and determine the direction of the equilibrium shift. A: $Q_c = 1.9 \times 10^{-7}$, shift left B: $Q_c = 6.45 \times 10^3$, shift right C: $Q_c = 4.6 \times 10^4$, none is favoured
25. Which of the statements defines the activity of a substance. A: degree of randomness

of a system B: a measure of its effective concentration under specified conditions. C: a measure of heat content of a substance

26. A system in which reactants and products are found in two or more phases is a. A: Phase equilibrium B: chemical equilibrium C: Heterogeneous equilibrium

27. One of the following is an example of heterogeneous equilibria. A: $C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

B: $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ C: $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^{-}(aq)$

28. A chemical reaction is said to be at equilibrium at. A: Minimal degree of freedom and maximum entropy B: Minimal degree of freedom and minimal entropy C: Minimal entropy and minimal degree of freedom

29. Predict the spontaneity of a chemical process if $\Delta G^\circ = 141.7$ and $K = 1.4 \times 10^{-25}$ at 298K. A: Spontaneous process B: Non spontaneous C: An equilibrium process

30. Estimate the boiling point of water given $\Delta H^\circ = 44.01 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = 118.3 \text{ J/K mol}$. A: 93.7°C B: 97.3°C C: 97.3 kJ/mol

31. Which of the following statements is true for the effect of catalyst on equilibrium. A: affects the rates of the reactions B: alter the equilibrium position C: catalyst remains in the reactants

32. Which of the following principles is used in predicting changes in equilibrium concentrations? A: Fritz Haber B: Le Chateliers C: Zeroth

33. The following reaction is at equilibrium. $C_2H_2(g) + 3F_2(g) \rightleftharpoons 2CF_3(g)$ How will the system respond if the volume is increased at constant temperature? A: The reaction will shift to the right. B: There will be no change to the equilibrium position C: The reaction will shift to the left

34. The following reaction is at equilibrium. $CF_4(g) + 2Br_2(g) \rightleftharpoons CF_2Br_2(g) + 2Br(g)$ $\Delta H = 424 \text{ kJ mol}^{-1}$ How will the system respond if the temperature is decreased? A: The reaction will shift to the left. B: The reaction will shift to the right. C: reaction stops

35. Consider this equation: $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$ Suppose the equation is rewritten as $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$ with an equilibrium constant K_c' . What is the relationship between K_c and K_c' ? A: $K_c = K_c'$ B: $K_c = (K_c')^2$ C: $K_c = \frac{1}{2}(K_c')$

36. Identify the INCORRECT statement below regarding chemical equilibrium. A: All chemical reactions are, in principle, reversible. B: Equilibrium is achieved when the concentrations of species become constant. C: Equilibrium is achieved when reactant and product concentrations are equal.

37. In which of the following reactions will the point of equilibrium shift to the left when the

pressure on the system is increased? A: $\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons \text{CO}_2\text{(g)}$
 B: $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$ C: $2\text{Mg(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{MgO(s)}$

38. What happens when a catalyst is added to a system at equilibrium? A: The reaction follows an alternative pathway of lower activation energy. B: The heat of reaction decreases. C: The potential energy of the reactants decreases.

39. The reaction $\text{A} \rightleftharpoons \text{B}$ has an equilibrium constant of $K = 10^{-4}$. Which of the following statements is always correct? A: The reaction will have 50% product B and 50% reactant A at equilibrium. B: The reaction is very favourable and will have mostly product B at equilibrium. C: The reaction is unfavourable and will not have very much product B at equilibrium.

40. Which of the following statements most accurately relates the properties of a liquid at room temperature with its vapour pressure? A: liquid low vapour pressure will probably have a high surface tension and a high boiling point. B: liquid low vapour pressure will probably have a low surface tension and a high boiling point. C: liquid high vapour pressure will probably have a low surface tension and a high boiling point.

41. For a reversible reaction, the equilibrium lies to the. A: Left B: Right C: middle

42. Which of the following is true for the composition of equilibrium mixture. If $\Delta G = 0$ and $K = 1$ the mixture is. A: Mostly products B: Mostly reactants C: Neither reactants nor products are favoured

43. Calculate the value of K at 298K for the following reaction $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$ Given $\Delta G = -32.96\text{KJmol}^{-1}$. A: 5.97×10^5 B: 13.3 C: -13.3

44. Of the following reactions, which of the reaction process is only spontaneous at high temperatures? A: $\Delta H +$, $\Delta S -$ B: $\Delta H +$, $\Delta S +$ C: $\Delta H -$, $\Delta S -$

45. The incomplete combustion of carbon is described by the following equation $2\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$ How does the spontaneity of this process depend upon temperature? A: spontaneous ($\Delta G < 0$) at all temperatures. B: Spontaneous process at high temperature C: Non spontaneous process

46. If ΔH is negative and ΔS is positive, this condition describes. A: Spontaneous process at low temperature B: Non spontaneous process C: Spontaneous process at all temperatures

47. A condition of a predation process describes an endothermic process with an increase in system entropy, ΔG will be negative if. A: $T\Delta S$ is greater than ΔH B: $T\Delta S$ is less than ΔH C: $T\Delta S = 0$

48. At 25°C, a reaction has a Gibb's free energy change of +45kJ. If the enthalpy change of the reaction is +35kJ, what is the entropy change of the reaction? A: -400JK B: -33.6JK C: 33.6JK

49. Energy can neither be created nor destroyed but can be converted from one form to other is inferred from. A: zeroth law of thermodynamic B: first law of thermodynamics C: second law to

thermodynamics

50. From the equation $2\text{NO}(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + \text{O}_2(\text{g})$

When the value of the reaction quotient before any reaction occurs is zero (0) at 25 °C, the concentration changes so that at equilibrium, $[\text{NO}] = 0.016 \text{ M}$ and

$[\text{N}_2] = 0.042 \text{ M}$. What is the value of the equilibrium constant for the reaction? A: 1.6×10^2 B: -1.6×10^5 C: 0.016

51. For the reaction, $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ the

concentrations at equilibrium are $[\text{SO}_2] = 0.90 \text{ M}$, $[\text{O}_2] = 0.35 \text{ M}$, and $[\text{SO}_3] = 1.1 \text{ M}$. What is the value of the equilibrium constant,

K_c ? A: $K_c = 3.4$ B: $K_c = 6.9$ C:

$K_c = 4.3$

52. A 1.00-L flask containing 0.0500 mol of $\text{NO}(\text{g})$, 0.0155 mol of $\text{Cl}_2(\text{g})$, and 0.500 mol of $\text{NOCl}(\text{g})$

$2\text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{NOCl}(\text{g})$ $K_c = 4.6 \times 10^4$

Calculate the reaction quotient and determine the direction of the equilibrium shift. A: $Q_c = 0$, shifts right B: $Q_c = 1.9 \times 10^{-7}$, shift

left C: $Q_c = 6.45 \times 10^3$, shift right

53. Which of the statements defines the activity of a substance. A: degree of randomness of a system B: a measure of its effective concentration under specified conditions. C: a measure of heat content of a substance

54. A system in which reactants and products are found in two or more phases is a. A: Phase equilibrium B: chemical equilibrium C: Heterogeneous equilibrium

55. The following factors affects the position of an equilibrium EXCEPT. A: Temperature B: Pressure C: Catalyst

56. Which one of the following statements regarding energy is false? A: The total energy in a chemical universe (a system and its surroundings) is constant B: Energy can be converted from one form to another C: The energy stored in chemical bonds is referred to as kinetic energy

57. First law of thermodynamics furnishes the relationship between. A: heat and work B: heat, work and properties of the system C: various properties of the system

58. In which direction does the transfer of energy as heat happen spontaneously? A: From cold to hot B: From hot to cold C: No changes occurs

59. Which one of the following statements best describes the enthalpy change of a reaction? A: The energy released when chemical bonds are formed during a chemical reaction B: The energy consumed when chemical bonds are broken during a chemical reaction C: The difference between the energy released by bond formation and the energy consumed by bond cleavage during a chemical reaction

60. Enthalpy is represented by which of the following symbols? A: H B: K C: S

61. Which of the following terms describes a reaction in which there is a net transfer of energy from a system to its surroundings? A: Exothermic B: Endothermic C: Spontaneous

62. Which of the following statements regarding the Gibbs free energy change for a reaction is

false? A: The Gibbs free energy change is the proportion of the enthalpy change of a reaction that is used to increase the entropy. B: If the Gibbs free energy change for a reaction is negative, the reaction happens spontaneously. C: The Gibbs free energy is represented by the symbol G

63. Energy is measured in which of the following units? A: Kelvin B: Joule C: Pascal

64. The formation of a chemical bond releases energy. True or false? A: True B: False C: All of the above

65. In case of spontaneous reaction, the reaction process favoured is. A: Forward B: Both direction C: Reverse

66. Which of the following is true for a closed system? A: mass entering = mass leaving B: mass does not enter or leave the system C: no reaction occur

67. When the value of K is greater than the I, then the reaction is. A: Forward reaction B: Backward reaction C: Spontaneous

68. ΔG for spontaneous reaction is. A: Negative B: Positive C: zero

69. Describe how matter and/or energy is redistributed when you empty a canister of compressed air into a room. A: greater and more uniform dispersal of matter B: much lesser dispersal of matter C: a reversible process

70. Describe how matter and/or energy is redistributed when you empty a canister of compressed air into a room. A: greater and more uniform dispersal of matter B: much lesser dispersal of matter C: a reversible process

71. Why do plastic materials tend to persist in the environment? A: Oxidation of plastics is fast and spontaneous B: Plastics are kinetically unstable C: Oxidation of plastics is slow and spontaneous

72. Predict the sign of the entropy change for the following processes.
 $C_6H_6(l) + 15/2 O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$

A: ΔS = positive B: ΔS = negative C: ΔS = 0

73. $NaNO_3(s) \rightarrow Na^+(aq) + NO_3^-(aq)$. A: ΔS = positive B: ΔS = negative C: ΔS = +/-

74. The freezing of liquid water. A: ΔS = positive B: ΔS = negative C: ΔS is = 0

75. Will Ice Spontaneously Melt? The entropy change for the process $H_2O(s) \rightarrow H_2O(l)$ Given the ΔS of the system and the surrounding as 22.1J/K and requires that the surrounding transfer 6.00kJ of heat to the system. Is the process spontaneous at -10.00 °C? Is it a spontaneous at +10.00 °C? A: Melting is spontaneous at 10 °C B: Melting is non spontaneous at 10 °C C: Entropy is equal to 0

76. Determine if liquid water will spontaneously freeze at the same temperatures.
 $H_2O(s) \rightarrow H_2O(l)$ Given the ΔS of the system and the surrounding as 22.1J/K and requires that the surrounding transfer 6.00kJ of heat to the system. Is the process spontaneous at -10.00 °C? Is it a spontaneous at +10.00 °C? A: Freeing is spontaneous at -10 °C B: Freeing is spontaneous at 10 °C C: Entropy is equal to 0

77. $S = k \ln \Delta$; $W = k \ln(1) = 0$
 The above equation represents. A: First law of thermodynamics B: Second law of thermodynamics C: Third law of thermodynamics
78. What is the difference between S° , and S°_{298} for a chemical change? A: S° standard enthalpy at 1 bar pressure B: ΔS°_{298} Standard change in entropy at room temperature C: S° Entropy of a perfect crystal at absolute zero
79. Which of the following options indicates that a chemical reaction is unfavorable? A: ΔH is negative B: ΔG is negative C: ΔS is negative
80. Consider the following reaction in a galvanic cell
 $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$
 Which if the following statements about the reaction is false? A: The cell potential is positive B: The equilibrium constant is greater than one C: Gibbs free energy is positive
81. The two ways in which a closed system can share energy with its surroundings are ----- and ----
 -- A: heat and work B: exothermic and endothermic C: hot and cold
82. If a system is at equilibrium, what is the relationship between ΔH and ΔS . A: $\Delta S = T \Delta H$ B: $\Delta S = T \Delta H$ C: $\Delta S = \Delta H/T$

1/19/2018 InfoMAS CHM111

[Log in Sign up](#)

[Home](#)

[About Us](#)

[Handouts](#)

[Latest Info](#)

[PastExams Q A](#)

[Welcome!](#)

[InfoMAS Get Q&A](#)

The Information Management and Admin System (InfoMAS)

is the initiative of students

from the SICT, FUTMinna. You

are encourage to use this past questions and

answers in addition to the course

material. The InfoMAS should not be held

responsible for any failure in the CA/Exam

resulting from over reliance on this Q&A in

the course of your preparation. It is at your

own discretion to use this material in part

or in totality for your preparation for the

test/exam

Select a Course to view

Select course to view

Search!

PAST QUESTION:

CHM 111

Chemistry.

According to the kinetic theory,
molecules of different gasses
at the same temperature
always have the same ____

Average density

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 1/12

1/19/2018 InfoMAS CHM111

Volume Log in Sign up

Pressure

Average kinetic energy

Doubling the volume of an
ideal gas at constant
temperature will

Double the pressure

Half the temperature

Half the pressure

Increase the pressure by halve

As solid elements melts, the
atom becomes ____ and they
have ____ attraction for one
another

Closer together, more

More separated, more

Larger greater

More separated, less

The tendency of an electrode
to release to or attract electron
from the surrounding
electrolyte solution at 298K
temperature 1atm pressure and
1ml/dm³ conc. Is known as

Electrolytic potential

Electrode potential

Standard electrolytic potential

Standard electrode potential

Consider the reaction 2AgNO_3

$+ \text{H}_2\text{S}(\text{g}) \rightleftharpoons \text{Ag}_2\text{S}(\text{s}) + 2\text{HNO}_3(\text{aq})$;

Which of the following affects
the value of the solubility
product?

Increase in temperature

The addition of aqueous

sodium sulphate

Addition of aqueous silver
nitrate

The presence of excess acid

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 2/12

1/19/2018 InfoMAS CHM111

The molar solubility of lead (II) chloride is $1.5 \times 10^{-2} \text{ mol/dm}^3$.
Log in Sign up

What is the solubility product
of the salt?

3.0×10^{-2}

3.0×10^{-4}

1.5×10^{-2}

1.4×10^{-5}

When ΔH is negative, the
reaction is said to be

Endothermic

Exothermic

Reversible

Catalytic

Which of the following
statement is true of an acid

Insane »

conjugate base?

Neutralized one another

Differ only by proton

Form salt and water only

Carry opposite charges

The pK_b of the base XOH, YOH
and ZOH are 2.70, 8.60 and 4.50
respectively. Arrange them in
order of increasing strength

XOH, YOH, ZOH

YOH, XOH, ZOH

XOH, ZOH, YOH

YOH, ZOH, XOH

Use the data below to answer
question 41 - 43

The following are the solubility
product of some sulphides in
mol/dm³

MnS = 2.5×10^{-10} , (ii) FeS = 1.0×10^{-10}
(iii) ZnS = 4.5×10^{-24} (iv)
PbS = 0.01mol/dm³.

Give that 1.3×10^{-19} mol/dm³ of
hydrogen sulphide is bubbled
through a solution containing

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 3/12

1/19/2018 InfoMAS CHM111

0.01mol/dm³ each of the Log in Sign up
respective cations.

The order in which the salt will
precipitate is

MnS, FeS, ZnS PbS

FeS, MnS, ZnS, PbS

PbS, ZnS FeS MnS

ZnS, FeS, PbS, MnS

The cation with the lowest
concentration is

Zn²⁺

Pb²⁺

Fe²⁺

Mn²⁺

The cation with the highest
concentration is

Zn²⁺

Pb²⁺

Fe²⁺

Mn²⁺

The ionic product in a
saturated aqueous solution at
a given temperature is given as

K_n

K_w

Ksp

Ka

When POH is less than 7.0 the
solution is said to be

Neutral

Alkaline

Acidic

Saturation

Which of the following
solutions is buffer system?

KNO₂/NHO₂

HCOOK/HCOOH

KH₂PO₄/H₃PO₂

KHSO₂/H₂SO₄

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 4/12

1/19/2018 InfoMAS CHM111

0.001mol/dm³ of Ba(OH)₂ has a Log in Sign up

pH of

7

11

9

10

Calculate the hydrolysis
constant of 0.1m NH₄Cl
solution in 1dm³ of water (kb of
NH₃ is 1.8 x 10⁻⁵mol/dm³)

5.6 x 10⁻⁹

5.6 x 10⁻¹⁰

5.6 x 10⁻²⁰

5.6 x 10⁻²¹

Which of the statements best,
describes an anode in an
electrochemical cell?

The electrode, at which
oxidation occurs

The electrode at which
electrons enter the cell

The electrode at which
reduction occurs

The electrode at which higher

deposition occurs

The pH of 0.01 HCl solutions

is?

1.0

2.0

3.0

4.0

The POH of 0.01 NaOH

solutions is

10

8

12

4

2

Which of the following can act

as bronsted acid?

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 5/12

1/19/2018 InfoMAS CHM111

BF3 Log in Sign up

HS

NH3

CO3

Lowry Bronsted defines acid

as

A proton acceptor

An electron acceptor

A proton donor

An electron donor

Calculate the concentration of

OH ion in a HCl solution whose

H⁺ is 1.3mol/dm³

7.7×10^{-17}

7.7×10^{-16}

7.7×10^{-15}

7.7×10^{-14}

Calculate the pH of a solution

whose hydrogen ion

concentration is 0.76mol/dm³

1.0

2.0

0.12

1.2

When a weak acid react with a strong base in a neutralization reaction, the resultant solution is

Basic

Neutral

Acidic

Normal

The weakest interparticle attraction exist between particles of ____ and the strongest interparticle attraction exist between particles of a ____

Liquid, gas

Gas, liquid

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 6/12

1/19/2018 InfoMAS CHM111

Solid, gas Log in Sign up

Gas, solid

Which of the following statement is not true about the effect of an increase in temperature on the distribution of molecular speeds in a gas?

The most probable speed increase

The fraction of the molecules with the most probable speed increase

The distribution becomes broader

The area under the distribution curve remains the same as under the lower temperature.

A reaction involving two different reactants can never be

Unimolecular reaction

First order reaction

Second order reaction

Enmolecular reaction

Energy conservation is the
basis of

Avogadro's hypothesis

Hess's law

Graham law of diffusion

Law of multiple proportion

The entropy of a system will
increase in the order

Gas > liquid > solid

Liquid > solid > gas

Solid > liquid > gas

None of the above

The property of substance that
relate to its ability to form new
substances are labeled

Physical

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 7/12

1/19/2018 InfoMAS CHM111

Chemical Log in Sign up

Extensive

Intensive

Catenation

An example of physical change
produce by addition of energy
is

Lighting of a match

Burning

Melting of ice

Condensation of steam

When a substance change and
still retains its original
properties, the change is called

Physical change

Endothermic change

Exothermic change

Chemical change

Which of the following is true?

If two gases under the same conditions have the same number of molecule then they must (i) have the same number of molecule (ii) occupy the same volume (iii) different number of molecules (iv) occupy different volume

I and II

II and III

III and IV

In thermo chemistry, the reference standard state for gases, liquid and solid is said to be at

25oC/1atm

0oC/10atm

1oC/60atm

1273K/10atm

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 8/12

1/19/2018 InfoMAS CHM111

The thermodynamic state function that is a measure of disorderliness is

Lavoisier

Dalton

Proust

Boyta

The salt resulting from the reaction of both strong acid and base is likely to be a/an

Acid salt

Basic salt

Neutral

No reaction

Use this question to answer the following next two questions

If the hydrogen ion of a solution is given as 0.0001mol/dm^3 what is the pH of the solution?

- 1
- 2
- 3
- 4
- 8

From the result obtain above we can conclude that the solution is

- Acidic
- Basic
- Neutral
- Natural

The pK_b of the base XOH, YOH and ZOH are 2.70, 8.60 and 4.50 respectively. Arrange them in order of increasing strength

- XOH, YOH, ZOH
- YOH, XOH, ZOH
- XOH, ZOH, YOH
- YOH, ZOH, XOH

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 9/12

1/19/2018 InfoMAS CHM111

Log in Sign up

Lowry Bronsted defines acid as

- A proton acceptor
- An electron acceptor
- A proton donor
- An electron donor

Doubling the volume of an ideal gas at constant temperature will

- Double the pressure
- Half the temperature
- Half the pressure
- Increase the pressure by halve

When a strong acid react with a weak base in a neutralization reaction, the resultant solution is

- Basic
- Neutral
- Acidic
- Normal

When a strong acid react with a strong base in a neutralization reaction, the resultant solution is

- Basic
- Neutral
- Acidic
- Normal

At equilibrium, the change in free energy of a reaction is

- Positive
- Zero
- Minimum
- Negative

The Free Energy of a Standard Solution at equilibrium is

- Positive

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 10/12

1/19/2018 InfoMAS CHM111

Zero Log in Sign up

- Minimum
- Negative

The unit of measurement for the solubility of a substance is_____

- mol-1
- mol dm³
- mol dm⁻³
- dm³/mol

Obtain K_c for the reaction H₂

+ O₂ ----->H₂O

$\frac{[H][O]}{[H_2O]}$

[H]₂[O]₂ / [H₂O]

[H₂O] / [H]₂[O]₂

None of the Above

What is a buffer solution?

Type your Answer here

Go Top | Home | FUTMininfo.hexat.com

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 11/12

1/19/2018 InfoMAS CHM111

Log in Sign up

<http://futmininfo.hexat.com/InfoMASCHM111.htm> 12/12

DEPARTMENT OF CHEMISTRY

FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

FIRST SEMESTER CONTINUOUS ASSESSMENT 2009/2010 SESSION

COURSE CODE: CHM 111 UNITS: 3

COURSE TITLE: PHYSICAL CHEMISTRY I

TIME ALLOWED: 30 minutes

INSTRUCTIONS: Answer all questions

Express the rate of the reaction in terms of oxygen concentration is

- A.
- B.
- C. *
- D.

In terms of vapour concentration, the rate expression of the reaction below is

- A.
- B. *
- C.
- D.

Study the reaction $2\text{H}_2(\text{g}) + 1/2\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$ if decreases by 0.23mol/L.s. What is the rate at which increase?

- 0.46 mol/L.s
- +0.46mol/L.s*
- 0.23mol/L.s
- 0.23mol/L.s

Which of these is a mathematical statement of Charles's law?

- A. $V \propto P$
- B. $V \propto 1/P$
- C. $V \propto T^*$
- D. $V = T$

The mass number of an element is 22 and the atomic number is 10.

The number of protons is

- A. 12
- B. 22
- *C. 10
- D 32.

The number of neutrons of the element are

- *A. 12
- B. 11
- C. 22
- D. 10

1

Use the diagram below to answer the following THIRTEEN questions

Z

U

Pressure (mmHg)

T

Y

V X

W

T(oC)

Y on the diagram represents

- A. Gas
- *B. Solid
- C. Liquid
- D. Vapour

What phase is represented by 'T'?

- A. Vapour
- B. Gas
- *C. Solid
- D. Liquid

At point 'V' the phases that co-exist are

- A. Solid and Gas only.
- B. Solid and liquid only.
- C. Gas, liquid and vapour only.
- *D. Liquid, solid and vapour only.

Only vapour is found at

- *A. X
- B. U

C. Y

D. W

Z is the

*A. Critical point

B. Boiling point

C. Melting point

D. Azeotropic point

From X to Y one witness

*A. A drop in temperature

B. No change in temperature

C. An increase in pressure

D. An increase in volume

The diagram represents the phase diagram for

A. Carbon (IV) oxide

B. Kerosene

*C. Water

D. Ethanol

2

Sublimation is possible while moving from ---- to ----

A. W - V

*B. Y - X

C. T - W

D. V - S

Sporting activities are commonly possible at

A. X and Y

B. W and U

C. X and U

*D. T and Y

Phases which coexist at ZV are

A. solid and vapour

B. liquid and solid

*C. liquid and vapour

D. solid, liquid and vapour

Phases which coexist at ZV and WV are

A. solid and vapour

B. liquid and solid

C. liquid and vapour

*D. solid, liquid and vapour

From the diagram with respect to Y the pressure – temperature relationship is

A. Constant Proportionality

- B. Reciprocal Proportionality
- C. Direct Proportionality.
- *D. Inverse Proportionality

Solid-liquid equilibrium is represented by

- A. UV
- *B. WV
- C. ZV
- D. XT

The rate expression for the reaction is

- A. *
- B.
- C.
- D.

The rate expression of a given reaction is . The negative sign means

- A. P is the product.
- B. P is decreasing *
- C. P is adding
- D. P is increasing

3

From the reaction , how fast is O₂ decreasing when NO reduces at the rate of $1.6 \times 10^{-5} \text{ mol/L.s}$

- 0.00008 mol/L.s*
- 0.00032 mol/L.s
- 0.0016mol/L.s
- 0.008mol/L.s

From the diagram below answer the THREE questions that follow

0.008mol/L.s

II

I

IV

III

t

The average rate of the reaction is represented by

- A. I
- B. II*
- C. III
- D. IV

The instantaneous rate of the reaction can be found at

- A. I

- B. II
- C. III*
- D. IV

The initial rate of the reaction is the slope of

- A. I*
- B. II
- C. III
- D. IV

If one of the species involved in a reaction is colored. Such reaction can be monitored by a

- A. Manometer
- B. Titrimetr
- C. Colorimeter*
- D. UV visible

If one of the species of a reaction is a gas --- method is suitable for monitoring such reaction.

- A. Manometer*
- B. Titrimetr
- C. Colorimeter
- D. pH meter

4

Hydrolysis of ethyl acetate can be monitored by

- A. titrimetry*
- B. colorimetry
- C. pH meter
- D. monmetry

It was observed that increase concentration of a specie in a chemical reaction had NO effect on the rate of

the reaction. The reaction is ---- order with respect to the specie

- A. zero*
- B. first
- C. second
- D. third

Doubling the concentration of a reactant doubles the rate of the reaction. The reaction is ---- order with

respect to the specie

- A. zero
- B. first*
- C. second
- D. third

Doubling the concentration of a reactant quadruples the rate of the reaction. The reaction is ---- order with

respect to the specie

- A. zero
- B. first
- C. second*
- D. third

It gives the number of molecule which takes part in a chemical reaction. What is it?

- A. Order
- B. Rate constant
- C. Molecularity*
- D. Rate

Which of the following may NOT be determined experimentally?

- A. Oder
- B. Rate constant
- C. Rate
- D. Molecularity*

The heat of a system measured at constant pressure may be taken as

- A. Enthalpy*
- B. Entropy
- C. Heat
- D. Work

From the reaction, which of the following is not true about the system?

- A. Volume is constant
- B. $P\Delta V = 0$
- C. Moles of reacts = moles of product
- D. $P\Delta V \neq 0$ *

5

A test tube feels cold, the process taken place in it is

- A. Endothermic*
- B. Exothermic
- C. Adiabatic
- D. Isochoric

Use the diagram below to answer the ELEVEN questions that follow;

T

R

Pressure (mmHg)

S

Q

O U

P

T(oC)

Only vapour is found at

- A. R
- *B. U
- C. Q
- D. O

Sublimation is possible when changing from --- to ---

- A. Q-S
- B. O-T
- *C. Q-U
- D. O-R

Sporting activities are commonly possible at

- A. Q and U
- *B. Q and S
- C. S and U
- D. O and S

From the diagram with respect to Q the pressure – temperature relationship is

- A. Direct Proportionality.
- *B. Inverse Proportionality
- C. Constant Proportionality
- D. Reciprocal Proportionality

T is the

- *A. Critical point
- B. Boiling point
- C. Melting point
- D. Azeotropic point

The diagram represents the phase diagram for

- A. Carbon (IV) oxide
- B. Kerosene
- *C. Water
- D. Ethanol

6

Q on the diagram represents

- A. Gas
- *B. Solid
- C. Liquid
- D. Vapour

What phase is represented by 'S'?

- A. Gas
- B. Solid
- *C. Vapour

D. Liquid

At point 'O' the phases that co-exist are

A. Solid and Gas only.

B. Solid and liquid only.

C. Gas, liquid and vapour only.

*D. Liquid, solid and vapour only.

Sporting activities are possible at Q during the

A. summer

B spring

*C. winter

D. autum

From U to Q one witnesses

A. An increase in pressure

B. An increase in volume

*C. A drop in temperature

D. No change in temperature

The energy involved when a substance melts is call

A. Activation energy

B. Heat of fission

C. Heat of fussion*

D. Heat of vaporization

Heat required to raise the temperature of a given mass of a substance by 1K is

A. Specific heat capacity

B. Heat capacity*

C. Heat of fussion

D. Heat of neutralization

How much heat is required to raise the temperature of a substance, X from 250C to 3000C? ($C = 48.37\text{J/K}$)

A. 4.838J

B. 26.51kj*

C. -48.38kj

D. 26.51J

When the three types of energy changes are arranged in order of DECREASING molar heat values, what is the correct order?

A. chemical, phase, nuclear

B. nuclear, phase, chemical

C. nuclear, chemical, phase*

D. phase, chemical, nuclear

The temperature remains constant as energy is added to a substance. How may the substance be changing?

7

- A. from a gas to a solid
- B. from a liquid to a solid
- C. from a liquid to a gas *
- D. in the amount of kinetic energy

A substance has a molar heat of combustion of -810.4 kJ/mol . When 0.285 mol of the substance is burned in a

calorimeter containing 8.60 kg of water, what is the increase of the water temperature?

- A. 0.156°C
- B. 6.41°C^*
- C. 7.89°C
- D. 12.8°C

The amount of 2.00 mol of a solid ionic compound was dissolved in 225 mL of water in an insulated container. The

initial temperature of the water was 20.0°C and the final temperature of the water was 47.8°C .

What is Heat of solution?

- A. -27.8 kJ/mol
- B. -26.2 kJ/mol
- C. -13.1 kJ/mol^*
- D. -6.26 kJ/mol

What does it mean if the ΔH value for a chemical reaction is positive?

- A. kinetic energy is increasing in the system
- B. potential energy is decreasing in the system
- C. products have less potential energy than reactants
- D. reactants have less potential energy than products. *

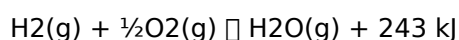
Living plants produce glucose in the process of photosynthesis according to this equation:



Is this reaction endothermic or exothermic, and is the value of ΔH° positive or negative?

- A. endothermic, positive *
- B. endothermic, negative
- C. exothermic, positive
- D. exothermic, negative

In order to produce 972 kJ of heat, how many grams of H_2 must burn?



- A. 0.250 g
- B. 4.04 g
- C. 8.08 g^*
- D. 16.0 g

Use the equations to answer the question.



What is the value of ΔH for the reaction $2A + BC \rightarrow A_2B + C$?

- A. +1084.8 kJ
- B. +650.2 kJ *
- C. -650.2 kJ
- D. -1084.8 kJ

Consider this reaction: $\text{LiH(s)} + 90.5 \text{ kJ} \rightarrow \text{Li(s)} + \frac{1}{2}\text{H}_2\text{(g)}$

8

What amount of heat is necessary to decompose 15.9 g of LiH into Li and H₂ at standard conditions?

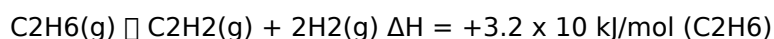
- A. 15.9 kJ
- B. 90.5 kJ
- C. 181 kJ *
- D. 1.44 MJ

During the production of a small amount of material, there is a large decrease in the temperature of the water in the

calorimeter. What is the ΔH for the reaction?

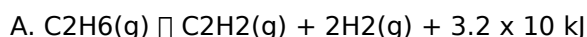
- A. large and positive *
- B. small and negative
- C. small and positive
- D. large and negative

2



What is an alternate way to write this equation?

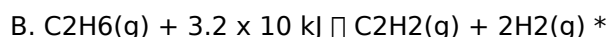
2



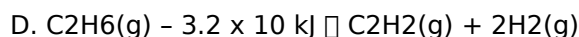
2



2



2



What is 4.18 J?

- A. The heat required to raise the temperature of one gram of water by one Celsius degree. *
- B. The heat required to raise the temperature of one mole of water by one Celsius degree.
- C. The heat required to raise the temperature of one gram of substance by one Celsius degree.
- D. The heat required to raise the temperature of one mole of substance by one Celsius degree.

The addition of 9.54 kJ of heat is required to raise the temperature of 225.0 g of a liquid hydrocarbon from 20.5°C to

45.0°C. What is the heat capacity of this hydrocarbon?

- A. 0.94 J/g•°C
- B. 1.73 J/g•°C*
- C. 1.88 J/g•°C
- D. 9.42 J/g•°C

If the molar heat of fusion of sodium is 2.63 kJ/mol, how much energy is needed to melt 180.0 g of solid sodium at its melting point?

- A. 2.63 kJ
- B. 2.93 kJ
- C. 20.6 kJ*
- D. 473 kJ

Which process is exothermic?

- A. the boiling of liquid nitrogen
- B. the sublimation of dry ice
- C. the freezing of water*
- D. the vaporization of water

If the heat of formation for the chemical reaction, $8\text{S(s)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{SO}_3\text{(g)}$, is -395.7 kJ/mol of SO_3 , how can the equation be rewritten to include the heat term?

- A. $8\text{S(s)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{SO}_3\text{(g)} + 3166 \text{ kJ}^*$
- B. $8\text{S(s)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{SO}_3\text{(g)} + 395.7 \text{ kJ}$
- C. $8\text{S(s)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{SO}_3\text{(g)} - 395.7 \text{ kJ}$

9

- D. $8\text{S(s)} + 12\text{O}_2\text{(g)} + 3166 \text{ kJ} \rightarrow 8\text{SO}_3\text{(g)}$

Which statement correctly describes the energy changes that take place when a solid changes to a liquid at constant temperature?

- A. The potential energy decreases, and the kinetic energy increases.*
- B. The potential energy remains constant, and the kinetic energy increases.
- C. The potential energy decreases, and the kinetic energy decreases.
- D. The potential energy increases, and the kinetic energy decreases.

Why does warm water have a more rapid rate of evaporation than cold water?

- A. It has a higher viscosity.
- B. Its molecules have a higher average kinetic energy.*
- C. Its molecules have more attraction for one another.
- D. More of its molecules have nearly the same kinetic energy.

If the heat of fusion of a substance is 20 kJ/mol, what heat is released when 1.0 mol of liquid at the melting point freezes?

- A. 10 kJ

- B. 20 kJ*
- C. 40 kJ
- D. 80 kJ

What is a positive molar heat of formation?

- A. The heat absorbed when one mole of compound is formed from its elements. *
- B. The heat released when one mole of compound is formed from its elements.
- C. The heat absorbed when one mole of elements is formed from the compound.
- D. The heat released when one mole of elements is formed from the compound.

What change is probably happening when energy is added to a substance, but the substance's temperature does not rise?

- A. gas to liquid
- B. gas to solid
- C. liquid to gas *
- D. liquid to solid

A sample of aluminum absorbed 9.86 J of heat and its temperature increased from 23.2°C to 30.5°C. What is the mass of the aluminum?

- A. 1.5 g *
- B. 8.1 g
- C. 65 g
- D. 72 g

At standard pressure, which is an example of a change in kinetic energy only?

- A. carbon dioxide cooling from -80°C to -100°C *
- B. molten aluminum solidifying at 660°C
- C. steam condensing at 100°C
- 7
- D. water decomposing above 1 x 10 °C

In order for 10.0 g of NH₄Cl to dissociate in water, 2.77 kJ of energy is required. What is the molar enthalpy of solution for NH₄Cl?

- A. 0.277 kJ/mol
- B. 2.77 kJ/mol
- C. 14.8 kJ/mol *
- D. 19.8 kJ/mol

Which statement is true?

- A. An endothermic reaction is characterized by a negative value of ΔH .
- 10
- B. An exothermic reaction transfers heat to the surroundings. *
- C. Heat is evolved when an endothermic reaction occurs.
- D. The reaction vessel cools when an exothermic reaction occurs.

Living plants produce glucose in the process of photosynthesis according to this equation:



Is this reaction endothermic or exothermic, and is the value of ΔH° positive or negative?

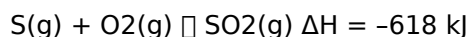
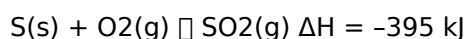
- A. endothermic, positive *
- B. endothermic, negative
- C. exothermic, positive
- D. exothermic, negative

Using the thermochemical equation: $2\text{Al}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) \Delta H = -1676 \text{ kJ}$

What is ΔH for the following reaction: $2\text{Al}_2\text{O}_3(\text{s}) \rightarrow 4\text{Al} + 3\text{O}_2(\text{g})$?

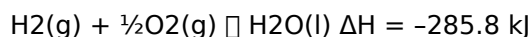
- A. -3352 kJ
- B. -838 kJ
- C. +838 kJ
- D. +3352 kJ *

What is the value of ΔH for the reaction, $\text{S}(\text{s}) \rightarrow \text{S}(\text{g})$?



- A. -1013 kJ
- B. -223 kJ
- C. + 223 kJ *
- D. +1013 kJ

What quantity of heat is evolved when 5.55 mol $\text{H}_2\text{O}(\text{l})$ is formed from the combustion of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$?



- A. 51.44 kJ
- B. 285.8 kJ
- C. 1586 kJ *
- D. 2297 kJ

Which processes are exothermic?

1. boiling water 2. freezing water 3. condensation of steam 4. melting ice

- A. 1 and 3
- B. 1 and 4
- C. 2 and 3 *
- D. 2 and 4

Which is correct for the process $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$?

- A. ΔH is negative and the process is endothermic.
- B. ΔH is positive and the process is endothermic.
- C. ΔH is negative and the process is exothermic. *
- D. ΔH is positive and the process is exothermic.

If 1.0 mol of ammonia reacts to form nitrogen and hydrogen,



what heat energy is associated with the reaction?

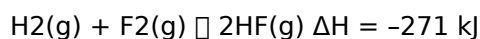
- A. 46 kJ and the reaction is endothermic *
- B. 92 kJ and the reaction is endothermic

11

- C. 46 kJ and the reaction is exothermic
- D. 92 kJ and the reaction is exothermic

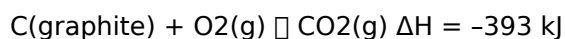
12

How might the reaction between $\text{H}_2(\text{g})$ and $\text{F}_2(\text{g})$ in a thermally insulated vessel be described?



- A. It is endothermic, and the temperature of the reaction system would fall.
- B. It is endothermic, and the temperature of the reaction system would rise.
- C. It is exothermic, and the temperature of the reaction system would fall.
- D. It is exothermic, and the temperature of the reaction system would rise. *

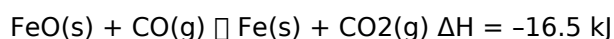
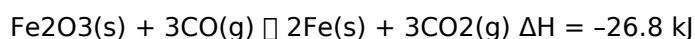
When graphite and diamond burn, the reactions can be represented by the thermochemical equations:



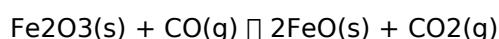
If 1.0 mol of graphite is converted into 1.0 mol of diamond under the same conditions, what is the heat energy change?

- A. 2 kJ of heat are absorbed from the surroundings. *
- B. 2 kJ of heat are given off to the surroundings.
- C. 788 kJ of heat are absorbed from the surroundings.
- D. 788 kJ of heat are given off to the surroundings.

Use the thermochemical equations:



to calculate ΔH for this reaction:

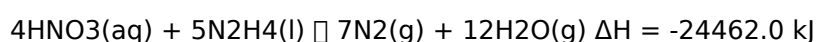


- A. -59.8 kJ
- B. -43.3 kJ
- C. -10.3 kJ
- D. +6.2 kJ *

Which one of the following processes is endothermic?

- A. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- B. $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$
- C. $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$
- D. $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$ *

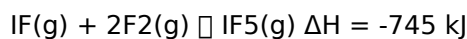
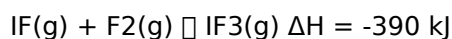
Which statement is true for the reaction between $\text{HNO}_3(\text{aq})$ and $\text{N}_2\text{H}_4(\text{l})$?



- A. 205.2 kJ are absorbed per mole of $\text{H}_2\text{O}(\text{g})$ used.

- C. 492.4 kJ are released per mole of $\text{N}_2\text{H}_4(\text{l})$ formed.
- B. 351.7 kJ are absorbed per mole of $\text{N}_2(\text{g})$ formed.
- D. 615.5 kJ are released per mole of $\text{HNO}_3(\text{aq})$ used. *

Given the two reactions below, what is the ΔH for the reaction, $\text{IF}_5(\text{g}) \rightarrow \text{IF}_3(\text{g}) + \text{F}_2(\text{g})$?



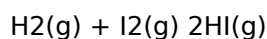
- A. -1135 kJ
- B. 35 kJ
- C. 355 kJ *
- D. 1135 kJ

For which of the following equations is the enthalpy change equal to ΔH°_f for the product?

- A. $2\text{Ca}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CaO}(\text{s})$
- B. $2\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g})$
- C. $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_4(\text{g})$
- D. $3\text{Mg}(\text{s}) + \text{N}_2(\text{g}) \rightarrow \text{Mg}_3\text{N}_2(\text{s})$ *

13

Calculate the ΔH for the following reaction using the bond energies given below.



Bond Energies: $\text{H}-\text{H} = 436 \text{ kJ/mol}$, $\text{I}-\text{I} = 151 \text{ kJ/mol}$, $\text{H}-\text{I} = 297 \text{ kJ/mol}$

- A. +290 kJ
- B. -290 kJ
- C. +7 kJ
- D. -7 kJ *

Which statement correctly describes an endothermic chemical reaction?

- A. The products have higher potential energy than the reactants, and ΔH is negative.
- B. The products have higher potential energy than the reactants, and the ΔH is positive. *
- C. The products have lower potential energy than the reactants, and the ΔH is negative.
- D. The products have lower potential energy than the reactants, and the ΔH is positive.

The following decomposition reaction may occur in an air bag. $2\text{NaN}_3(\text{s}) \rightarrow 3\text{N}_2(\text{g}) + 2\text{Na}(\text{s})$ $\Delta H = -43.5 \text{ kJ}$

What is the heat of formation, ΔH_f , for NaN_3 ?

- A. -43.5 kJ
- B. -21.8 kJ
- C. 21.8 kJ *
- D. 43.5 kJ

10

A small sample released $2.0 \times 10^3 \text{ kJ}$ of energy while undergoing a change. What type of change most likely occurred?

- A. chemical
- B. molecular

C. nuclear *

D. physical

Use the diagram below to answer the following THIRTEEN questions

H

Y

Pressure (mmHg)

X

Z

F G

E

T(oC)

Sporting activities are commonly possible at

A. X and Y

*B. Z and X

C. X and G

D. H and E

From the diagram with respect to Z the pressure – temperature relationship is

A. Constant Proportionality

*B. Inverse Proportionality

C. Direct Proportionality.

D. Reciprocal Proportionality

solid-liquid equilibrium is represented by

A. FY

14

B. FH

C. GF

*D. EF

The diagram represents the phase diagram for

A. Carbon (IV) oxide

*B. Water

C. Kerosene

D. Ethanol

Z on the diagram represents

A. Gas

B. Liquid

*C. Solid

D. Vapour

What phase is represented by 'G'?

A. Vapour

*B. Solid

C. Gas

D. Liquid

At point 'F' the phases that co-exist are

A. Solid and Gas only.

*B. Liquid, solid and vapour only.

C. Gas, liquid and vapour only.

D.. Solid and liquid only.

Only vapour is found at

A. Y

B. U

*C. G

D. W

Phases which coexist at HF are

*A. liquid and vapour

B. liquid and solid

C. solid and vapour

D. solid, liquid and vapour

Phases which coexist at HF and FE are

A. solid and vapour

*B. solid, liquid and vapour

C. liquid and vapour

D. liquid and solid

H is the

A. Melting point

B. Boiling point

*C. Critical point

D. Azeotropic point

From G to Z one witness

A. No change in temperature

*B. A drop in temperature

C. An increase in pressure

D. An increase in volume

15

Sublimation is possible while moving from ---- to ----

A. E - F

B. F -YS

C. G - X

*D. G - Z

Which of these is a mathematical statement of Charle's law?

- A. $V \propto P$
- B. $V \propto T$
- *C. $V \propto 1/P$
- D. $V = T$

The mass number of an element is 22 and the atomic number is 10.

The number of protons are

- *A. 10
- B. 22
- C. 12
- D 32.

The number of neutrons of the element are

- A. 22
- B. 11
- *C. 12
- D. 10

For the ELEVEN questions that follow, Use the diagram below

S

U

Pressure (mmHg)

T

V

R X

W

T(oC)

S, is the

- *A. Critical point
- B. Boiling point
- C. Melting point
- D. Azeotropic point

Sporting activities are possible at V during the

- *A. winter
- B spring
- C. summer
- D. autum

from X to V one witnesses

- A. An increase in pressure
- 16
- *B. A drop in temperature
- C. An increase in volume
- D. No change in temperature

The diagram represents the phase diagram for

- A. Carbon (IV) oxide
- B. Kerosene
- *C. Water
- D. Ethanol

V on the diagram represents

- A. Gas
- *B. Solid
- C. Liquid
- D. Vapour

What phase is represented by 'T'?

- A. Gas
- B. Solid
- *C. Vapour
- D. Liquid

At point 'R' the phases that co-exist are

- A. Solid and Gas only.
- B. Solid and liquid only.
- C. Gas, liquid and vapour only.
- *D. Liquid, solid and vapour only.

Only vapour is found at

- A. R
- *B. U
- C. Q
- D. R

Sublimation is possible when changing from --- to ---

- A. V-T
- B. R-S
- *C. V-X
- D. R-U

Sporting activities are commonly possible at

- A. V and X
- *B. V and T
- C. T and X
- D. R and S

From the diagram with respect to V the pressure – temperature relationship is

- A. Direct Proportionality.
- *B. Inverse Proportionality
- C. Constant Proportionality
- D. Reciprocal Proportionality

The rate expression for the reaction is

A. *

17

B.

C.

D.

The rate expression of a given reaction is . The negative sign means

A. P is the product.

B. P is decreasing *

C. P is adding

D. P is increasing

Express the rate of the reaction in terms of oxygen concentration

In terms of vapour concentration, the rate expression of the reaction below is

Study the reaction if decreases by 0.23mol/L.s. what is the rate at which increase?

A. -0.46 mol/L.s

B. +0.46mol/L.s*

C. 0.23mol/L.s

D. -0.23mol/L.s

From the reaction , how fast is O₂ decreasing when NO reduces at the rate of 1.6x10⁻⁵ mol/L.s

A. 0.00008 mol/L.s*

B. 0.00032 mol/L.s

C. 0.0016mol/L.s

D. 0.008mol/L.s

From the diagram below to answer the TWO questions that follow;

18

I

II

III

IV

The instantaneous rate of the reaction can be found at

A. IV* t

B. III

C. II

D. I

The average rate of the reaction is represented by

A. IV

B. III

C. II

D. I*

The initial rate of the reaction is the slope of

A. IV

B. III

C. II

D. I*

If one of the species involved in a reaction is colored. Such reaction can be monitored by

A. Colorimeter*

B. Titrimetr

C. Manometer

D. UV visible

If one of the species of a reaction is a gas --- method is suitable for monitoring such reaction.

A Manometer*

B Titrimetr

C Colorimeter

D pH meter

Hydrolysis of can be monitored by

A. titrimetry*

B. colorimetry

C. pH meter

D. monometry titrimetry

It was observed that increase concentration of a specie in a chemical reaction had no effect on the rate of the

reaction. The reaction is ---- order with respect to the specie

A. Colorimeter first

B. second

C. third

19

D. zero*

Doubling the concentration of a reactant doubles the rate of the reaction. The reaction is ---- order with respect

to the specie

A. first *

B. second

C. third

D. zero

Doubling the concentration of a reactant quadruples the rate of the reaction. The reaction is ---- order with

respect to the specie

A. first

- B. second*
- C. third
- D. zero

It gives the number of molecule which takes part in a chemical reaction. What is it?

- A. Rate constant
- B. Molecularity*
- C. Rate
- D. Order

Which of the following may not be determined experimentally?

- A. Molecularity*
- B. Rate constant
- C. Rate
- D. Oder

The heat of a system measured at constant pressure may be taken as

- A. Entropy
- B. Heat
- C. Work
- D. Enthalpy*

From the reaction which of the following is NOT true about the system?

- A. Volume is constant
- B. $P\Delta V \neq 0$ *
- C. $P\Delta V = 0$
- D. Moles of reacts = moles of product

A test tube feels cold, the process taken place is

- A. Exothermic
- B. Adiabatic
- C. Endothermic*
- D. Isochoric

The energy involved when a substance melts is call

- A. Activation energy
- B. Heat of fussion*

20

- C. Heat of fission
- D. Heat of vaporization

Heat required to raise the temperature of a give mass of a substance by 1K is

- A. Specific heat capacity
- B. Heat of fussion
- C. Heat of neutralization
- D. Heat capacity*

How much heat is required to raise the temperature of a substance, X from 250C to 3000C? (C = 48.38J/K)

- A. 483.8J
- B. -48.38kj
- C. -26.51J
- D. 26.51kj*

When the three types of energy changes are arranged in order of DECREASING molar heat values, what is

the correct order?

- A. chemical, phase, nuclear
- B. nuclear, phase, chemical
- C. nuclear, chemical, phase*
- D. phase, chemical, nuclear

The temperature remains constant as energy is added to a substance. How may the substance be changing?

- A. from a liquid to a gas*
- B. from a liquid to a solid
- C. from a gas to a solid
- D. in the amount of kinetic energy

A substance has a molar heat of combustion of -810.4 kJ/mol. When 0.285 mol of the substance is burned

in a calorimeter containing 8.60 kg of water, what is the increase of the water temperature?

- A. 0.156°C
- B. 12.8°C
- C. 7.89°C
- D. 6.41°C *

The addition of 9.54 kJ of heat is required to raise the temperature of 225.0 g of a liquid hydrocarbon from

20.5°C to 45.0°C. What is the heat capacity of this hydrocarbon?

- A. 1.73 J/g•°C *
- B. 0.94 J/g•°C
- C. 1.88 J/g•°C
- D. 9.42 J/g•°C

If the molar heat of fusion of sodium is 2.63 kJ/mol, how much energy is needed to melt 180.0 g of solid

sodium at its melting point?

- A. 20.6 kJ *
- B. 2.93 kJ
- C. 2.63 kJ
- D. 473 kJ

Which process is exothermic?

- A. the freezing of water *
- C. the sublimation of dry ice
- B the boiling of liquid nitrogen
- D. the vaporization of water

21

If the heat of formation for the chemical reaction, $8\text{S(s)} + 12\text{O}_2(\text{g}) \rightarrow 8\text{SO}_3(\text{g})$, is -395.7 kJ/mol of SO_3 ,

how can the equation be rewritten to include the heat term?

- A. $8\text{S(s)} + 12\text{O}_2(\text{g}) \rightarrow 8\text{SO}_3(\text{g}) - 395.7 \text{ kJ}$
- A. $8\text{S(s)} + 12\text{O}_2(\text{g}) \rightarrow 8\text{SO}_3(\text{g}) + 395.7 \text{ kJ}$
- C. $8\text{S(s)} + 12\text{O}_2(\text{g}) \rightarrow 8\text{SO}_3(\text{g}) + 3166 \text{ kJ}$ *
- D. $8\text{S(s)} + 12\text{O}_2(\text{g}) + 3166 \text{ kJ} \rightarrow 8\text{SO}_3(\text{g})$

Which statement correctly describes the energy changes that take place when a solid changes to a liquid at

constant temperature?

- A. The potential energy increases, and the kinetic energy decreases.
- B. The potential energy remains constant, and the kinetic energy increases.
- C. The potential energy decreases, and the kinetic energy decreases.
- D. The potential energy decreases, and the kinetic energy increases.*

Why does warm water have a more rapid rate of evaporation than cold water?

- A. It has a higher viscosity.
- B. More of its molecules have nearly the same kinetic energy.
- C. Its molecules have more attraction for one another.
- D. Its molecules have a higher average kinetic energy.*

If the heat of fusion of a substance is 20 kJ/mol , what heat is released when 1.0 mol of liquid at the melting

point freezes?

- A. 10 kJ
- B. 80 kJ
- C. 40 kJ
- D. 20 kJ *

The amount of 2.00 mol of a solid ionic compound was dissolved in 225 mL of water in an insulated container. The initial temperature of the water was 20.0°C and the final temperature of the water was 47.8°C .

What is Heat of solution?

- A. -27.8 kJ/mol
- B. -26.2 kJ/mol
- C. -13.1 kJ/mol *
- D. -6.26 kJ/mol

What does it mean if the ΔH value for a chemical reaction is positive?

- A. kinetic energy is increasing in the system

- B. reactants have less potential energy than products. *
- C. products have less potential energy than reactants
- D. potential energy is decreasing in the system

Living plants produce glucose in the process of photosynthesis according to this equation:

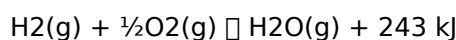


Is this reaction endothermic or exothermic, and is the value of ΔH° positive or negative?

- A. exothermic, positive
- B. endothermic, negative
- C. endothermic, positive *
- D. exothermic, negative

In order to produce 972 kJ of heat, how many grams of H_2 must burn?

22



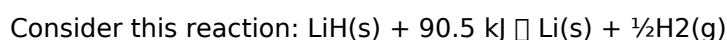
- A. 8.08 g *
- B. 4.04 g
- C. 0.250 g
- D. 16.0 g

Use the equations to answer the question.



What is the value of ΔH for the reaction $2\text{A} + \text{BC} \rightarrow \text{A}_2\text{B} + \text{C}$?

- A. +1084.8 kJ
- B. -1084.8 kJ
- C. -650.2 kJ
- D. +650.2 kJ *



What amount of heat is necessary to decompose 15.9 g of LiH into Li and H_2 at standard conditions?

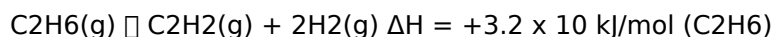
- A. 181 kJ *
- B. 90.5 kJ
- C. 15.9 kJ
- D. 1.44 MJ

During the production of a small amount of material, there is a large decrease in the temperature of the water

in the calorimeter. What is the ΔH for the reaction?

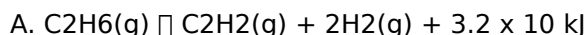
- A. small and positive
- B. small and negative
- C. large and positive *
- D. large and negative

2

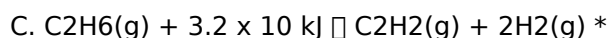


What is an alternate way to write this equation?

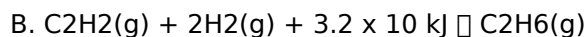
2



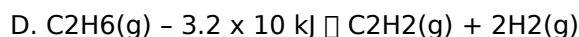
2



2



2



What is 4.18 J?

A. The heat required to raise the temperature of one gram of substance by one Celsius degree

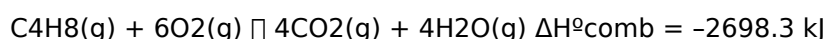
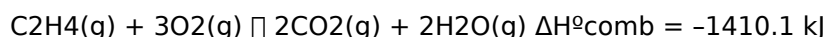
B. The heat required to raise the temperature of one mole of water by one Celsius degree.

C. The heat required to raise the temperature of one gram of water by one Celsius degree.*

D. The heat required to raise the temperature of one mole of substance by one Celsius degree.

23

Use these thermochemical equations to answer the question.



What is the heat of the reaction, $2\text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_4\text{H}_8(\text{g})$?

A. -5518.5 kJ

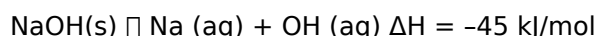
B. -121.9 kJ *

C. -1288.3 kJ

D. -3986.5 kJ

What can be concluded from this thermochemical equation?

+ -



A. Sodium and hydroxide ions have more potential energy than solid sodium hydroxide.

B. Solid sodium hydroxide has less potential energy than aqueous sodium hydroxide.

C. The temperature of the water would rise as solid sodium hydroxide dissolves.*

D. The crystallization of sodium hydroxide is an exothermic process.

What is one result of a solid-liquid phase change?

A. A change in a substance's potential energy. *

C. A change in a substance's mass.

B. A change in a substance's kinetic energy.

D. A change in a substance's chemical bonding

A sample of aluminum absorbed 9.86 J of heat and its temperature increased from 23.2°C to 30.5°C. What is the mass

of the aluminum?

A. 65 g

- B. 8.1 g
- C. 1.5 g *
- D. 72 g

At standard pressure, which is an example of a change in kinetic energy only?

- A. molten aluminum solidifying at 660°C
- B. carbon dioxide cooling from -80°C to -100°C *
- C. steam condensing at 100°C
- 7
- D. water decomposing above 1 x 10 °C

In order for 10.0 g of NH₄Cl to dissociate in water, 2.77 kJ of energy is required. What is the molar enthalpy of solution for NH₄Cl?

- A. 0.277 kJ/mol
- B. 2.77 kJ/mol
- C. 19.8 kJ/mol
- D. 14.8 kJ/mol *

Which statement is true?

- A. An exothermic reaction transfers heat to the surroundings. *
- B. An endothermic reaction is characterized by a negative value of ΔH.
- C. Heat is evolved when an endothermic reaction occurs.
- D. The reaction vessel cools when an exothermic reaction occurs.

Living plants produce glucose in the process of photosynthesis according to this equation:



Is this reaction endothermic or exothermic, and is the value of ΔH° positive or negative?

- A. endothermic, negative
- B. endothermic, positive *
- C. exothermic, positive
- D. exothermic, negative

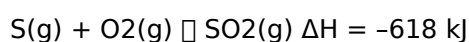
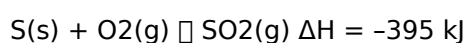
24

Using the thermochemical equation: $2\text{Al}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow \text{Al}_2\text{O}_3(\text{s})$ ΔH = -1676 kJ

What is ΔH for the following reaction: $2\text{Al}_2\text{O}_3(\text{s}) \rightarrow 4\text{Al} + 3\text{O}_2(\text{g})$?

- A. +3352 kJ *
- B. -838 kJ
- C. +838 kJ
- D. -3352 kJ

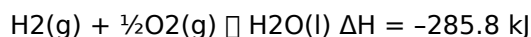
What is the value of ΔH for the reaction, $\text{S}(\text{s}) \rightarrow \text{S}(\text{g})$?



- A. -1013 kJ
- B. + 223 kJ *

- C. -223 kJ
- D. +1013 kJ

What quantity of heat is evolved when 5.55 mol H₂O(l) is formed from the combustion of H₂(g) and O₂(g)?



- A. 51.44 kJ
- B. 285.8 kJ
- C. 2297 kJ
- D. 1586 kJ *

Which processes are exothermic?

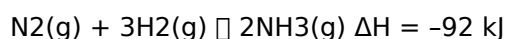
1. boiling water 2. freezing water 3. condensation of steam 4. melting ice

- A. 1 and 3
- B. 2 and 3 *
- C. 1 and 4
- D. 2 and 4

Which is correct for the process H₂O(g) → H₂O(l)?

- A. ΔH is negative and the process is endothermic.
- C. ΔH is positive and the process is endothermic.
- B. ΔH is positive and the process is exothermic
- D.. ΔH is negative and the process is exothermic. *

If 1.0 mol of ammonia reacts to form nitrogen and hydrogen,

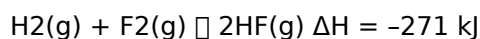


what heat energy is associated with the reaction?

- A. 46 kJ and the reaction is exothermic
- B. 92 kJ and the reaction is endothermic
- C. 46 kJ and the reaction is endothermic *
- D. 92 kJ and the reaction is exothermic

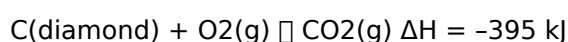
25

How might the reaction between H₂(g) and F₂(g) in a thermally insulated vessel be described?



- A. It is endothermic, and the temperature of the reaction system would fall.
- B. It is endothermic, and the temperature of the reaction system would rise.
- C. It is exothermic, and the temperature of the reaction system would fall.
- D. It is exothermic, and the temperature of the reaction system would rise. *

When graphite and diamond burn, the reactions can be represented by the thermochemical equations:

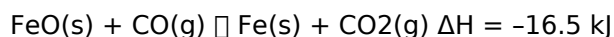
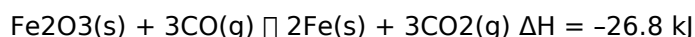


If 1.0 mol of graphite is converted into 1.0 mol of diamond under the same conditions, what is the heat energy change?

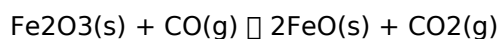
- A. 788 kJ of heat are absorbed from the surroundings.

- B. 2 kJ of heat are given off to the surroundings.
- C. 2 kJ of heat are absorbed from the surroundings. *
- D. 788 kJ of heat are given off to the surroundings.

Use the thermochemical equations:



to calculate ΔH for this reaction:

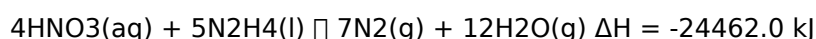


- A. -59.8 kJ
- B. +6.2 kJ *
- C. -10.3 kJ
- D. -43.3 kJ

Which one of the following processes is endothermic?

- A. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- B. $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$ *
- C. $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$
- D. $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$

Which statement is true for the reaction between $\text{HNO}_3(\text{aq})$ and $\text{N}_2\text{H}_4(\text{l})$?



- A. 205.2 kJ are absorbed per mole of $\text{H}_2\text{O}(\text{g})$ used.
- B. 615.5 kJ are released per mole of $\text{HNO}_3(\text{aq})$ used.*
- C. 351.7 kJ are absorbed per mole of $\text{N}_2(\text{g})$ formed.
- D. 492.4 kJ are released per mole of $\text{N}_2\text{H}_4(\text{l})$ formed.

Given the two reactions below, what is the ΔH for the reaction, $\text{IF}_5(\text{g}) \rightarrow \text{IF}_3(\text{g}) + \text{F}_2(\text{g})$?



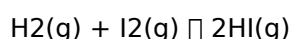
- A. 355 kJ *
- B. 35 kJ
- C. -1135 kJ
- D. 1135 kJ

For which of the following equations is the enthalpy change equal to ΔH°_f for the product?

- A. $2\text{Ca}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CaO}(\text{s})$
- B. $2\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g})$
- C. $3\text{Mg}(\text{s}) + \text{N}_2(\text{g}) \rightarrow \text{Mg}_3\text{N}_2(\text{s})$ *
- D. $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_4(\text{g})$

26

Calculate the ΔH for the following reaction using the bond energies given below.



Bond Energies: $\text{H}-\text{H} = 436 \text{ kJ/mol}$, $\text{I}-\text{I} = 151 \text{ kJ/mol}$, $\text{H}-\text{I} = 297 \text{ kJ/mol}$

- A. +290 kJ

- B. -7 kJ *
- C. +7 kJ
- D. -290 kJ

Which statement correctly describes an endothermic chemical reaction?

- A. The products have higher potential energy than the reactants, and ΔH is negative.
- B. The products have lower potential energy than the reactants, and the ΔH is positive.
- C. The products have lower potential energy than the reactants, and the ΔH is negative.
- D. The products have higher potential energy than the reactants, and the ΔH is positive. *

The following decomposition reaction may occur in an air bag. $2\text{NaN}_3(\text{s}) \rightarrow 3\text{N}_2(\text{g}) + 2\text{Na}(\text{s})$ $\Delta H = -43.5 \text{ kJ}$

What is the heat of formation, ΔH_f , for NaN_3 ?

- A. 21.8 kJ *
- B. -21.8 kJ
- C. -43.5 kJ
- D. 43.5 kJ

10

A small sample released $2.0 \times 10 \text{ kJ}$ of energy while undergoing a change. What type of change most likely occurred?

- A. chemical
- B. nuclear *
- C. molecular
- D. physical

Which is true for an exothermic reaction?

- A. The ΔH is positive.
- B. The products have less potential energy than the reactants. *
- C. The reactants have more kinetic energy than the products.
- D. The reactants are below the products in the potential energy diagram.

When $\text{NH}_4\text{Cl}(\text{s})$ is added to water, the resulting solution feels cool to the touch. Which best describes this observation?

- A. Heat is released as $\text{NH}_4\text{Cl}(\text{s})$ dissolves.
+ -
- B. $\text{NH}_4\text{Cl}(\text{s}) \rightarrow \text{NH}_4(\text{aq}) + \text{Cl}(\text{aq})$ $\Delta H = -33.6 \text{ kJ}$
+ -
- C. $\text{NH}_4\text{Cl}(\text{s}) + 33.6 \text{ kJ} \rightarrow \text{NH}_4(\text{aq}) + \text{Cl}(\text{aq})$ *
- D. $\text{NH}_4\text{Cl}(\text{s}) \rightarrow \text{NH}_4(\text{aq}) + \text{Cl}(\text{aq}) + 33.6 \text{ kJ}$

What characteristic(s) of water change(s) when its temperature is raised from 10°C to 110°C at standard pressure?

- A. intramolecular bonding only
- B. kinetic energy only

- C. both kinetic and potential energy *
- D. potential energy only

If 41.2 kJ of energy is needed to raise the temperature of 170.0 g of a liquid from 18.8°C to 65.2°C, what is the specific heat capacity of the solution?

- A. 4.19 J/g•°C
- B. 12.0 J/g•°C
- C. 11.2 J/g•°C
- D. 5.22 J/g•°C *

27

What change is probably happening when energy is added to a substance, but the substance's temperature does not rise?

- A. liquid to gas *
- B. gas to solid
- C. gas to liquid
- D. liquid to solid

What occurs when solid A (50°C) is placed in contact with solid B (80°C)?

- A. Heat energy flows from A to B as the average kinetic energy of the particles in A decreases.
- B. Heat energy flows from B to A as the average kinetic energy of the particles in B decreases. *
- C. Heat energy flows from A to B as the average kinetic energy of the particles in A increase.
- D. Heat energy flows from B to A as the average kinetic energy of the particles in B increases.

At what pressure is the temperature of water be 100 oC?

- (a) 760 atm
- (b) 1 mmHg
- (c) 273 atm
- (d) 760 mmHg*.

The FIVE questions below are based on this statement ;

A sample of argon gas occupies 105ml at 0.871atm. At constant temperature, the final pressure is 26.3

kPa.

What is volume (V1) in liter (L)?

- (a) 0.015L
- (b) 0.105L*
- (c) 0.150L
- (d) 1.05L.

The volume (V1) in cm³ is

- (a) 105cm³*
- (b) 1.05 cm³
- (c) 10.5 cm³
- (d) 0.105 cm³

The final pressure (P_2) in Pa is

- (a) 2630Pa
- (b) 26300Pa*
- (c) 263Pa
- (d) 26.300Pa.

The value for the pressure (P_2) in atm is

- (a) 2.596atm
- (b) 25.956atm
- (c) 259.6atm
- (d) 0.260atm*

Volume (V_2) is

- (a) 0.352L*
- (b) 0.0352L
- (c) 0.0035L
- (d) 3.52L

28

The followings are example of gaseous diatomic molecules except

- (a)Molecular hydrogen
- (b) Molecular nitrogen
- (c) Molecular chlorine
- (d) Ozone*.

----- are the most compressible of the states of matter.

- (a) Solids
- (b) Liquids
- (c) Gases*
- (d) Fluids.

Examples of monoatomic gas is

- (a) Ozone
- (b) Chlorine
- (c) Helium *
- (d) Hydrogen

A substance that is normally in the gaseous state at normal atmospheric conditions is

- (a) Gas*
- (b) Solid
- (c) Liquid
- (d) Vapor

----- is the gaseous form of any substance that is a liquid or a solid at normal temperature and pressure

- (a) Gas
- (b) Vapor *

- (c) Ozone
- (d) Ideal gases

The molecules of every gas do occupy some volume and exert some attractive or repulsive forces on each other.

- (a) TRUE *
- (b) FALSE

What is the instrument used to measure atmospheric pressure?

- (a) Monometer
- (b) Thermometer
- (c) Atmospheric meter
- (d) Barometer.*

The devices use to measure the pressure of gases other than the atmosphere is known as

- (a) Manometer *
- (b) Thermometer
- (c) Atmospheric meter
- (d) Barometer.

1 atm is equivalent to -----Pa.

- (a) 1.0325×10^2 Pa
- (b) 1.01325×10^5 Pa *

29

- (c) 1.01325×10^4 Pa
- (d) 1.0325×10^3 Pa.

The mathematical expression for Boyle's law is

- (a) $P/v = k$
- (b) $V/P = k$
- (c) $PT = k$
- (d) $PV = k^*$

The sum of all the mole fraction is unity.

- (a) TRUE *
- (b) FALSE

In Dalton's law of partial pressure, the gases are mixed at constant temperature and pressure

- (a) TRUE
- (b) FALSE*

Use the statement below to answer the TWO questions below.

An engineer pumps air at 0°C into a nearly designed piston-cylinder assembly. The initial volume (V_1) measure 6.83cm^3 while the final volume (v_2) is 9.75cm^3 .

The initial temperature (T_1) in Kelvin is given as

- (a) -273k

- (b) 273 ok
- (c) 273 k*
- (d) O k.

The value for temperature (T2) in Kelvin is

- (a) 191.24 k
- (b) 390 k *
- (c) 0.244 k
- (d) 0.0026 k.

The distance that a molecule moves through before colliding with another molecules is known as the -----

- (a) Average distance
- (b) Free path*
- (c) Mean free path
- (d) Average free path.

The larger the number of molecules, the larger the mean free path

- (a) TRUE
- (b) FALSE*

----- is the temperature above which a gas cannot be liquefied and below which a continuous increase of pressure result in the liquefaction of gas.

- (a) Liquefaction temperature
- (b) Critical temperature*
- (c) Thermal temperature
- (d) Absolute temperature.

30

The average distance transversed by a molecule between two successive collisions is referred to as

- (a) Average free path
- (c) Mean free path*
- (c) Free path
- (d) Collision distance.

Molecules of the same gas have the same mass, shape and size but does not differ from molecules of another gas.

- (a) TRUE
- (b) FALSE*.

Graham's law of diffusion can be represented as

- (a) $R_1/R_2 = \sqrt{d_1/d_2}$
- (b) $R_1/R_2 = \sqrt{M_1/M_2}$
- (c) $R_1/R_2 = \sqrt{d_2/d_1}$ *
- (d) $R_1/R_2 = \sqrt{M_1/M_2}$

Calculate the volume (in liters) occupied by 2.12 moles of nitric oxide at 6.54 atm and 760c.

$R = 0.0821 \text{ L. atm/k. mol}$

- (a) 9.92L
- (b) 9.29L *
- (c) 0.0033L
- (d) 88.39L

Calculate the ratio of the diffusion rates of helium and methane (CH_4).

$C = 12.01 \text{ g/mol}$. $H = 1.008 \text{ g/mol}$ and $\text{He} = 4.003 \text{ g/mol}$.

- (a) 2.02
- (b) 0.499
- (c) 2.002*
- (d) 4.007

Use the statement below to answer the following TWO questions

The total pressure exerted by a mixture of two gases 1 and 2 is 76.2 atm. The number of moles of the two gases are 0.0075 and 0.0040 respectively.

The partial pressure of the first gas is

- (a) 116.8 atm
- (b) 0.57 atm
- (c) 0.88 atm
- (d) 49.7 atm*

The second gas have the partial pressure to be

- (a) 0.3048 atm
- (b) 26.5 atm *
- (c) 19050 atm
- (d) 76.196 atm.

The law that supported the mixture of gases at constant pressure and temperature is known as

- (a) Avogadro's law
- (b) Gay- Lussac's law

31

- (c) Amagat's law*
- (d) Charle's law.

The mean free path is much longer in the liquid phase than in the gas phase.

- (a) TRUE
- (b) FALSE*

The pressure of a gas on any surface is the result of continuous bombardment on the surface by its molecules.

- (a) TRUE*
- (b) FALSE

The regions outside the boundaries of the system are termed as

- (a) Universe

- (b) Borders
- (c) Regionals
- (d) Surroundings*

A system where it is possible to exchange both energy and matter with its surroundings is known as

- (a) Closed system
- (b) Open system*
- (c) Isolated system
- (d) partial system

All are the variables that determine a complete system except

- (a) Energy *
- (b) Temperature
- (c) Pressures
- (d) volume.

First law of thermodynamics can also be regarded as

- (a) Law of conservation of matter
- (b) Law of conservation of system
- (c) Law of conservation of energy *
- (d) Law of conservation of universe

----- are those properties which depend upon the amount of the substance(s) present in the system.

- (a) Intensive properties
- (b) Extensive properties*
- (c) Isolated properties
- (d) Thermodynamics properties

----- is a system that can only exchange energy with its surroundings and the amount of matter in the system is not changed

- (a) Open system
- (b) Isolated system
- (c) Thermodynamics system
- (d) Closed system *

All the followings are examples of intensive properties except

32

- (a) Mass *
- (b) Pressure
- (c) Temperature
- (d) Viscosity

The properties that alter with the quantity of matter present in the systems are called

- (a) Intensive properties
- (b) Extensive properties *

(c) Thermodynamics properties

(d) Isolated properties

All the followings are example of extensive properties except

(a) Pressure *

(b) Mass

(c) Volume

(d) Heat capacity.

----- is a system where there is no transfer of either energy or mass with the surroundings.

(a) Open system

(b) Closed system

(c) Isolated system *

(d) Thermodynamics system

The properties which are independent of the amount of substance(s) present in a system are known as

(a) Extensive properties

(b) Thermodynamics properties

(c) Physical properties

(d) Intensive properties *

A gas can be liquefied by applying high temperature and decrease pressure.

(a) TRUE

(b) FALSE*

A liquid when placed in a container will partially evaporate and the molecules of the vapour would exert a definite pressure on the liquid

(a)TRUE*

(b) FALSE

The property common to both gases and liquids is

(a)Viscosity *

(b) System

(c) Surrounding

(d) Fluid

All the following are examples of metallic solids except

(a)NaF *

(b) Na

(c) Cu

(d) Au

A process where there is no exchange of heat between the system and its surrounding is known as

(a)Adiabatic process *

33

(b) Isothermal process

(c) Isobaric process

(d) Open process

----- is the science of crystals which deals with the study of their development and growth, their external form, internal structure and physical properties.

(a) Symmetry

(b) Crystallography *

(c) Crystalline

(d) Crystal properties.

What occurs when solid A (50°C) is placed in contact with solid B (80°C)?

A. Heat energy flows from A to B as the average kinetic energy of the particles in A decreases.

B. Heat energy flows from A to B as the average kinetic energy of the particles in A increase.

C. Heat energy flows from B to A as the average kinetic energy of the particles in B decreases.*

D. Heat energy flows from B to A as the average kinetic energy of the particles in B increases.

What characteristic(s) of water change(s) when its temperature is raised from 10°C to 110°C at standard pressure?

A. both kinetic and potential energy *

C. kinetic energy only

B. intramolecular bonding only

D. potential energy only

If 41.2 kJ of energy is needed to raise the temperature of 170.0 g of a liquid from 18.8°C to 65.2°C, what is the specific

heat capacity of the solution?

A. 4.19 J/g•°C

B. 5.22 J/g•°C *

C. 11.2 J/g•°C

D. 12.0 J/g•°C

What happens to the water in a calorimeter when an exothermic reaction occurs in it?

A. It absorbs heat, and a drop in temperature is observed.

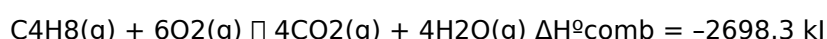
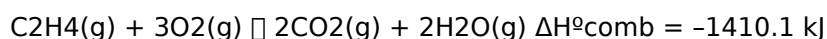
B. It absorbs heat, and a rise in temperature is observed.*

C. It releases heat, and a drop in temperature is observed.

D. It releases heat, and a rise in temperature is observed.

34

Use these thermochemical equations to answer the question.



What is the heat of the reaction, $2\text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_4\text{H}_8(\text{g})$?

A. -5518.5 kJ

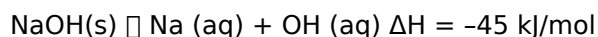
B. -3986.5 kJ

C. -1288.3 kJ

D. -121.9 kJ *

What can be concluded from this thermochemical equation?

+ -



- A. Sodium and hydroxide ions have more potential energy than solid sodium hydroxide.
- B. Solid sodium hydroxide has less potential energy than aqueous sodium hydroxide.
- C. The crystallization of sodium hydroxide is an exothermic process.
- D. The temperature of the water would rise as solid sodium hydroxide dissolves. *

What is one result of a solid-liquid phase change?

- A. A change in a substance's chemical bonding.
- B. A change in a substance's mass.
- C. A change in a substance's kinetic energy.
- D. A change in a substance's potential energy. *

What happens to the water in a calorimeter when an exothermic reaction occurs in it?

- A. It absorbs heat, and a drop in temperature is observed.
- B. It releases heat, and a drop in temperature is observed.
- C. It absorbs heat, and a rise in temperature is observed. *
- D. It releases heat, and a rise in temperature is observed.

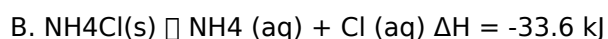
Which is true for an exothermic reaction?

- A. The products have less potential energy than the reactants. *
- B. The ΔH is positive.
- C. The reactants have more kinetic energy than the products.
- D. The reactants are below the products in the potential energy diagram.

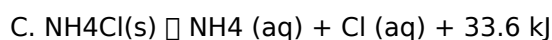
When $\text{NH}_4\text{Cl(s)}$ is added to water, the resulting solution feels cool to the touch. Which best describes this observation?

- A. Heat is released as $\text{NH}_4\text{Cl(s)}$ dissolves.

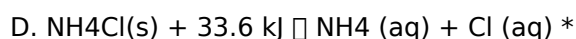
+ -



+ -



+ -



What is a positive molar heat of formation?

- A. The heat absorbed when one mole of elements is formed from the compound.
- B. The heat released when one mole of compound is formed from its elements.
- C. The heat absorbed when one mole of compound is formed from its elements. *
- D. The heat released when one mole of elements is formed from the compound.

35

CHM 111

SCANNED PAST QUESTIONS, FUTMINNA

Downloaded free from ujuzi.com.ng

This ebook has been made available for free download for all Students by the uJuzi-Project sponsored by Sterling Bank Plc.

CHM 111

SCANNED PAST QUESTIONS

SCANNED PAST QUESTIONS Chemistry

SCANNED PAST QUESTIONS Chemistry

This etextBook has been digitally prepared by the Content Management Team at ujuzi.com.ng under the leadership of Ms Nelly Tomilola. Thus, this ebook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it.

ujuzi.com.ng is an Online Educational Content distribution and delivery platform managed and owned by Aspire Solutions and Innovations.

Our aim is that students should be able to go through university without having to pay for textbooks. We hope to achieve this by publishing free and openly available etextBooks.

How is this possible

We finance our textbooks with a low number of high quality advertisements. We have set a 15% advertising limit per book. This limit has been set together with professors, business professionals, and student organizations in order to ensure the whole experience is in line with the high quality level of our books.

Other platforms under the uJuzi bouquet are;

SS

-www.ujuzi-tv.com is an online Educational TV station that'll feature some of the best Educational and academic video content.

-www.ujuzi.com.ng/college is an online free courses college for all. We envision a future where everyone has access to a world-class education that has so far been available to a select few.

Signed

Nelly Tomilola

Content Developer

www.ujuzi.com.ng

P.S

uJuzi is Pronounced "You-Ju-Zi" and its Swahili for Knowledge